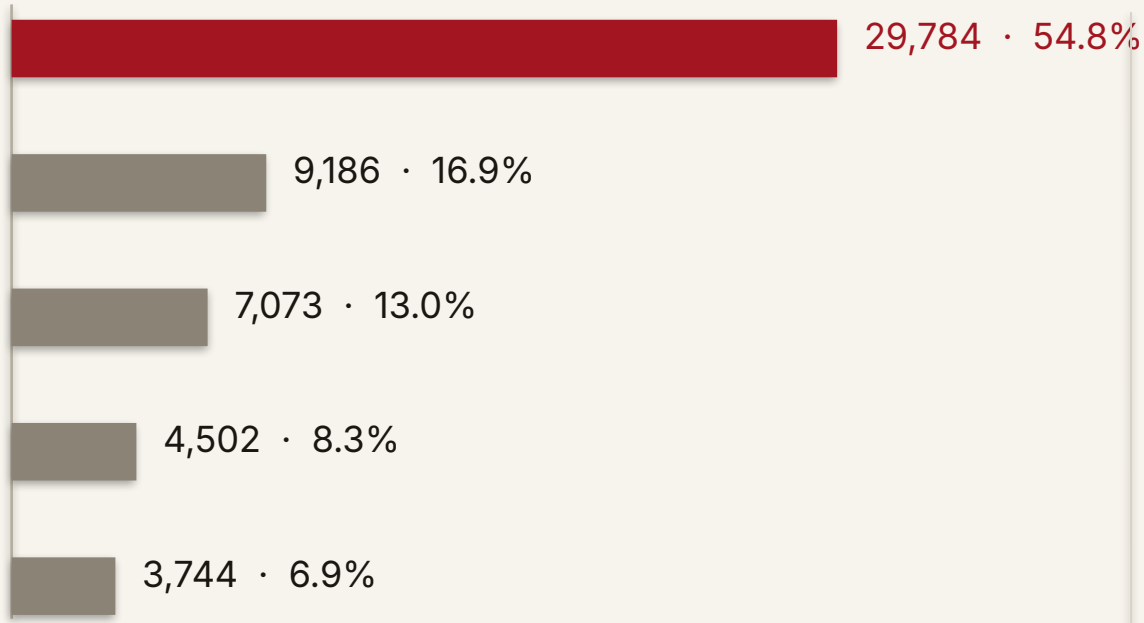
Framing, Sentiment, and Agenda-Setting Patterns: A Computational Analysis of News Coverage Across Five Major Outlets

Shruti Bahuguna

Graphic Era (Deemed to be) University, Dehradun

Seven months of news, five outlets

One shared dataset to compare how five very different outlets cover the same world · July 2025 – January 2026



54,289 articles in total

The Guardian is over half of all articles — so we re-checked every key result on a balanced sample (3,744 per outlet), and they held.

The five span the political spectrum, plus an international voice.

Only five, on purpose — every article gets the full analysis.

Seven months separates ongoing stories from one-week spikes.

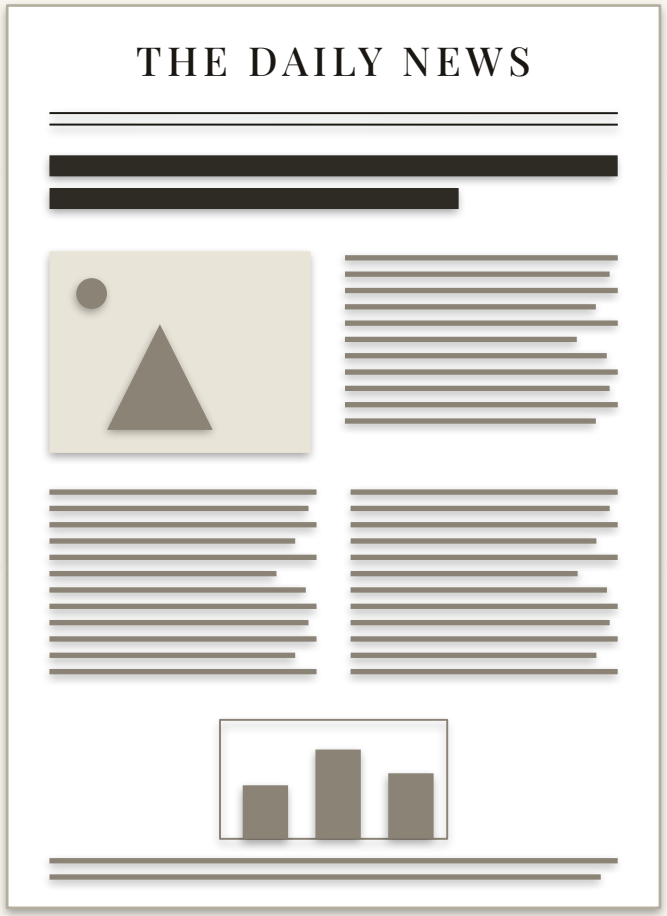
The pipeline works on any outlet with public feeds — more sources are future work.

We asked four questions of the same 54,289 articles

Most studies ask one of them. Asked together, the answers can be compared.

What gets covered?
topic classification

Who appears in it?
named entities



How negative is it?
sentiment score

Which words recur?
vocabulary profile

Four questions, four standard models

No custom training — published, publicly available models, run automatically over all 54,289 articles

every one of the 54,289 articles

What gets covered?	BART-large MNLI	assigns one of 11 topic categories to every article
How negative is it?	RoBERTa-large	scores each article positive, neutral or negative
Who appears in it?	spaCy transformer	finds every person named and merges name variants
Which words recur?	TF-IDF	weighs words one outlet uses far more than the others

IN NUMBERS

0.87
mean confidence

−1 to +1
tone scale

111,774
unique people

per outlet
vocabulary signature

Why these techniques? The tools are standard on purpose — we are measuring the coverage, and accepted instruments keep the focus on the data.

MNLI = Multi-Genre Natural Language Inference · TF-IDF = term frequency–inverse document frequency

We asked four questions. Here's what came back.

Three answers describe the coverage itself — one describes how we measure it.

1 · What gets covered? → Politics dominates everywhere — **12 politics stories for every science story**.

2 · How negative is it? → All five outlets skew negative; politics coverage is **4.4× more negative** than science.
And the way you measure it makes a person seem 37% more negative.

3 · Who appears in it? → One person, out of **111,774 unique people**, received **112,473 mentions** — nearly ten times anyone else.

4 · Which words recur? → Each outlet keeps a distinct vocabulary — **its own fingerprint**.

This is a measurement of coverage patterns — what is covered, who appears, in what tone. It is not a left-versus-right bias score.

ANSWER 1 OF 4 · WHAT GETS COVERED?

Politics gets 12× the coverage of science

Politics & Government: 25.2% of all articles · Science & Technology: 2.1%

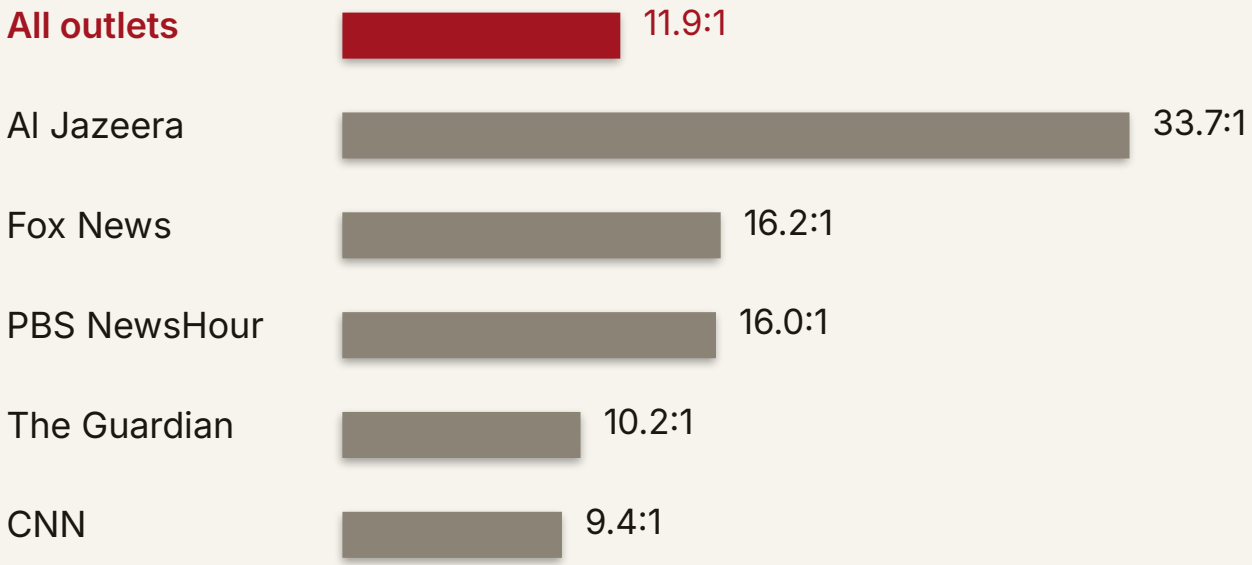
11.9 : 1

politics stories for every science story



12 politics 1 science

POLITICS-TO-SCIENCE RATIO



Each outlet also owns a beat: Al Jazeera — international affairs (a third of its output) · Fox News — crime (17% vs 9% average) · The Guardian — entertainment and sport (79% of all entertainment coverage)

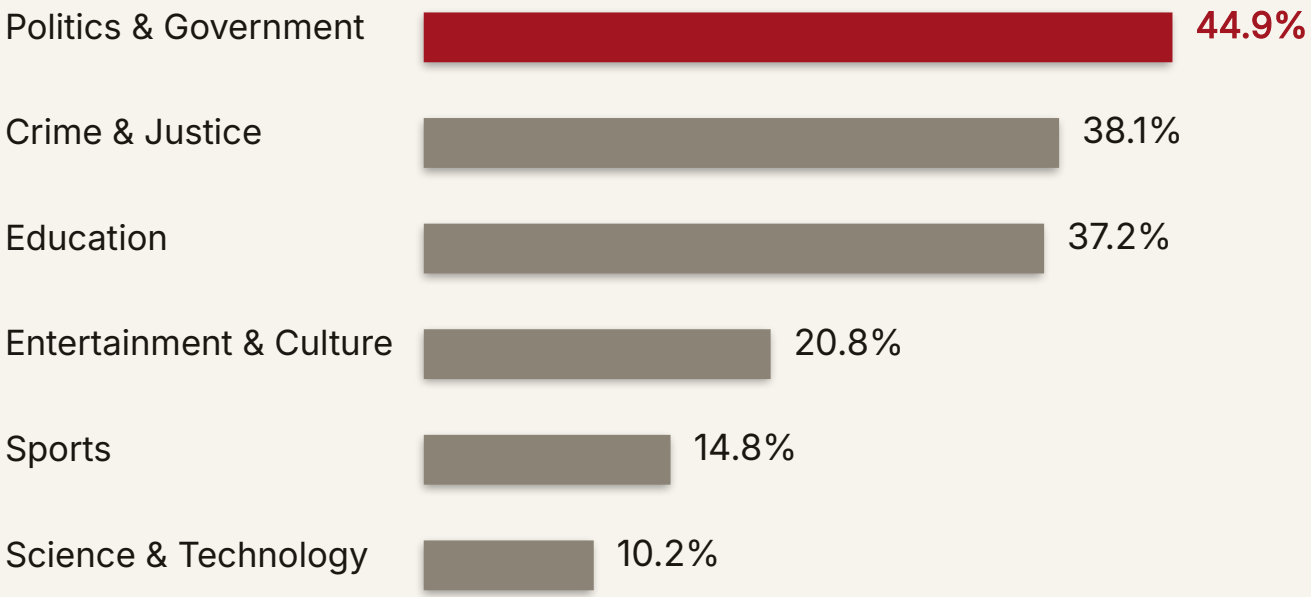
Topic labels from the zero-shot BART classifier · mean confidence 0.87 across all 54,289 articles

ANSWER 2 OF 4 · HOW NEGATIVE IS IT?

Tone follows the topic

Politics coverage is 4.4× more negative than science — and outlets across the spectrum are equally negative

SHARE OF ARTICLES SCORED NEGATIVE, BY TOPIC



AVERAGE TONE, BY LEANING GROUP

−1 = all negative · +1 = all positive

Center	−0.26
Center-left	−0.29
Right	−0.29

Every group averages a negative tone, and the largest gap between any two is 0.03 — too small to matter.

WHAT THIS MEANS

The news runs negative because of what it covers — political and civic stories — and outlets across the spectrum share the pattern.

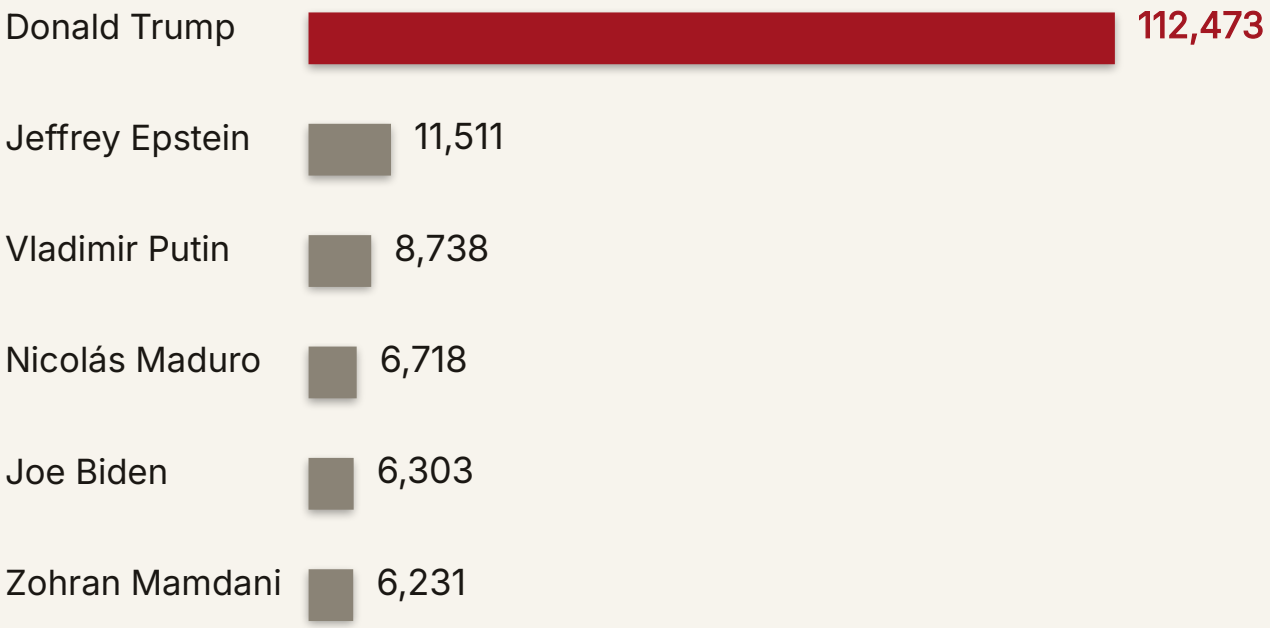
Headline tone matches article tone · tone scored by RoBERTa-large · six largest of 11 topics shown

ANSWER 3 OF 4 · WHO APPEARS IN IT?

Ten times more attention than anyone else

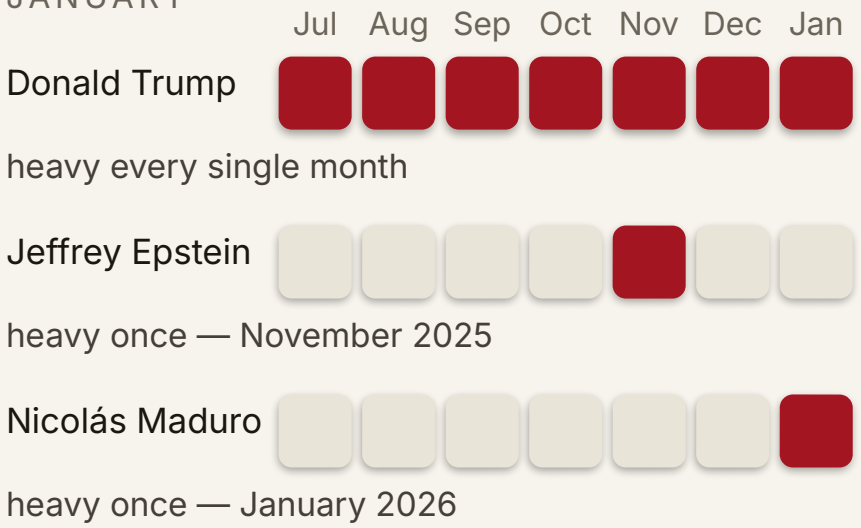
Donald Trump: 112,473 mentions — 9.8× Jeffrey Epstein, the next most-mentioned person

TOTAL MENTIONS, JULY 2025 – JANUARY 2026



Chi-squared test: could this just reflect how much each outlet publishes? No — the gap far exceeds what volume predicts ($p < 0.001$).

HEAVY-MENTION MONTHS, JULY → JANUARY



one square = one month · filled = heavy mentions that month

WHAT THIS MEANS

One person is a permanent fixture of the news; everyone else is a passing story.

ANSWER 4 OF 4 · WHICH WORDS RECUR?

Each outlet has its own vocabulary

The words each outlet uses far more often than the others do — its fingerprint (top TF-IDF terms)

The Guardian center-left	CNN + PBS NewsHour center / center-left	Al Jazeera international	Fox News right
uk	white house	israel	law enforcement
world	federal	gaza	president donald
prime minister	administration	military	fox news digital
human rights	justice department	prime minister	
far right	supreme court	united nations	

WHAT THIS MEANS

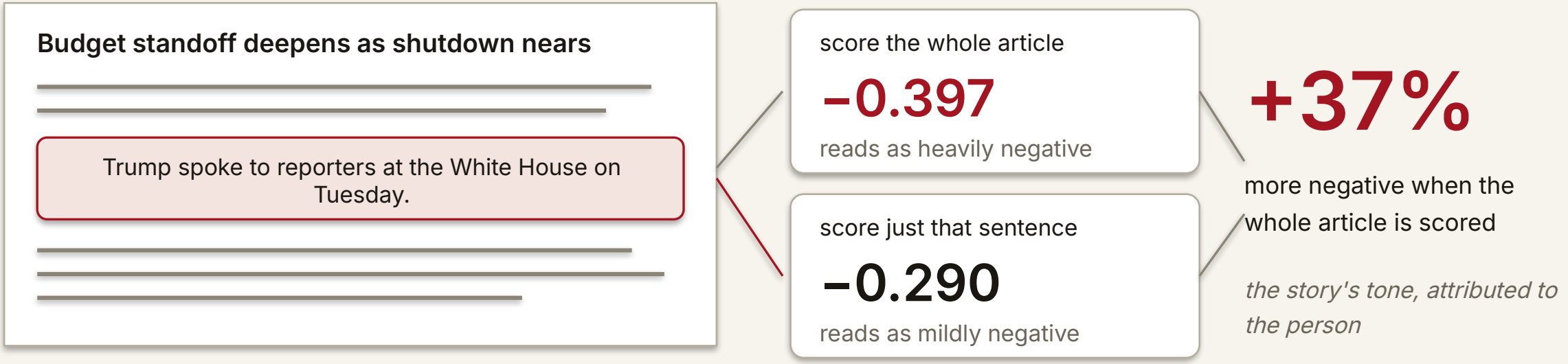
An outlet's editorial mandate is visible in its word choices — and CNN and PBS share theirs.

Distinctive words from TF-IDF: frequent in one outlet's articles, rare in the rest of the dataset

THE TONE ANSWER, REVISITED · HOW WE MEASURED IT

How you measure changes what you find

The experiment: score the same Trump mentions two ways — the whole article, or just the sentence that names him



one article about a budget fight — one sentence names him

And inside articles labeled negative, the sentences naming a person look like this:



about 7 of every 10 are neutral themselves (64.8%)

WHAT THIS MEANS

Score whole articles to judge a person and you overstate their negativity by about a third — score the sentence that names them.

Why trust the gap? We compared the two scores mention by mention — the whole-article score comes out more negative far too consistently to be chance (Wilcoxon signed-rank test, p ≈ 0), and the same happens for every top-10 figure.

CONCLUSIONS · LIMITATIONS · FUTURE WORK

What we found, what's new, what's next

11.9 : 1
politics vs science coverage

4.4×
political vs science negativity

9.8×
one person vs the next

+37%
negativity inflation from article-level scoring

WHAT'S NEW

— One corpus, all four dimensions — the answers can be compared.

— The 37% measurement inflation — quantified for the first time.

— A reproducible pipeline and benchmark numbers for the field.

WHAT IT CAN'T TELL YOU YET

— Five outlets, seven months — one election-cycle window.

— Topic labels are model judgments, not human-validated.

— Political leaning is labeled per outlet, not per article.

Next: more outlets and languages, longer windows — and sentence-level sentiment as the default when judging people.

Thank you — questions welcome.

Shruti Bahuguna · Graphic Era University
shrutibahuguna011@gmail.com